1MT495 Current Issues in Applied International Economics: New Trade Theory in International Trade

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Outline

Disclaimer: How to Use This Material

- **Scope:** My goal is to provide a broad overview of different models and approaches in modern international economics.
- **Focus:** We will emphasize the main ideas, core concepts, and ways of thinking, rather than an exhaustive dive into every mathematical detail. *(This is not a pure mathematics course.)*
- **Acknowledgement:** This material synthesizes and adapts lecture notes and core ideas from leading scholars in the field.
- **Sources:** I will share the original sources at the end of each lecture. Please consult them to explore topics in greater depth.

Traditional Trade Theories: Foundations and Limits

- **Comparative Advantage (Ricardo):** Trade arises solely from differences in **labor productivity** between countries.
- **Heckscher-Ohlin Model (H-O):** Trade is explained by differences in **factor endowments** (e.g., capital abundance vs. labor abundance).
- **Limitations of Traditional Models:**
  - Assume **perfect competition and constant returns to scale (CRS)**.
  - Predict trade only between dissimilar countries (inter-industry trade).
  - Fail to explain significant **intra-industry trade** (ITT) between similar, developed economies (e.g., US-EU).

The Need for New Trade Theory (NTT)

- **Empirical Reality:** Much of world trade is **intra-industry** (countries both import and export similar goods, e.g., cars).
- **Differentiated Products:** Traditional models cannot explain trade in **differentiated products** (different brands of cars, phones) without relying on comparative advantage.
- **Market Structure:** Emergence of large **multinational corporations** and the reality of increasing returns to scale.

NTT vs. Intertemporal Trade

- **Intertemporal Approach:** Explains trade across *time* (e.g., borrowing/lending, trade in assets).
- **New Trade Theory (NTT):** Explains trade between *similar countries* (horizontal intra-industry trade).
- **The Core Challenge:** Both approaches fundamentally challenge the classical view that trade arises only from differences in productivity, endowments, or time preferences.

Definition and Core Mechanisms

- **Origins:** Developed in the late 1970s and 1980s by Paul Krugman, Elhanan Helpman, and others.
- **Core Concepts:** Incorporates three key concepts:
  1. **Increasing Returns to Scale (IRS) / Economies of Scale.**
  2. **Imperfect Competition** (specifically, Monopolistic Competition).
  3. **Product Differentiation** (Consumer "love of variety").
- **Main Explanations:** Explains why countries trade similar goods and how trade creates welfare gains through scale efficiencies and greater consumer variety, even without underlying differences in factor endowments or technology.
- **Gravity Link:** Provides the theoretical foundation for the **Gravity Model of Trade** (explaining how trade flows are proportional to country economic size and negatively impacted by trade barriers, like distance).

Intra-Industry Trade in NTT

- Unlike traditional models, introduces intra-industry trade between similar countries.
- Countries specialize in different varieties of similar goods.
- Rationale for trade: Economies of scale and preference for variety, not comparative advantage or factor proportions.

Key Concepts

Economies of Scale

Production costs decrease as output increases, leading to larger firms and market concentration.

Imperfect Competition

Monopolistic competition: Many firms produce differentiated products, each with some market power.

Product Differentiation

Consumers prefer variety; firms differentiate products to capture market segments.

Case Study: Germany and Japan Automobiles

- **Initial Conditions:** Germany and Japan are similar in technology and factor endowments. Traditional models predict minimal trade.
- **Autarky (No Trade):** Limited domestic market size leads to:
  - Small-scale production per firm.
  - High average costs and prices.
  - Limited consumer choice (Germans only buy BMW/Mercedes; Japanese only Toyota/Honda).
- **With Trade:** The market expands for all firms to size $2L$.
  - Firms exploit greater scale, which reduces average costs and prices.
  - Consumers gain access to greater variety (Germans buy Toyota/Honda; Japanese buy BMW/Mercedes).
- **Conclusion:** Trade is driven by variety preference and scale efficiencies, not comparative advantage.

Krugman's Model (1980): Overview

- Two countries, identical in size and preferences.
- Firms produce differentiated goods under monopolistic competition with economies of scale.
- Economies of scale lead to specialization.
- Trade allows access to more varieties, increasing welfare.

Key Insight

Trade occurs even without comparative advantage, driven by economies of scale and product variety.

Assumptions of the Core Krugman Model

- **Technology:** Increasing returns to scale in production ($l = f + c \cdot q$).
- **Market Structure:** Monopolistic competition (firms face downward-sloping demand).
- **Preferences:** Differentiated products (Constant Elasticity of Substitution - CES), implying a **"love of variety".**
- **Countries:** Symmetric in terms of endowments (Labor L) and technology.
- **Trade Costs:** Initially, no transport costs for simplicity.

Krugman's Model: Consumer Preferences (CES)

Consumers maximize utility subject to their budget constraint.

$$U = \left(\int_0^n c_i^{(\sigma-1)/\sigma} \, di \right)^{\sigma/(\sigma-1)}$$

- $\sigma > 1$: Elasticity of substitution between varieties.
- **Love of Variety:** Utility (U) increases with the total number of varieties (n).
- **Demand:** Demand for each variety exhibits a constant elasticity of σ.

Derivation of Demand from CES (1/2)

The consumer solves:

$$\max U = \left(\int_0^n c_i^{(\sigma-1)/\sigma} \, di \right)^{\sigma/(\sigma-1)}$$

s.t. $\int_0^n p_i c_i \, di = Y$ (income/expenditure).
Using Lagrange multipliers or two-stage budgeting, the expenditure on variety i is:

$$e_i = p_i c_i = \left(\frac{p_i}{P} \right)^{1-\sigma} Y$$

where $P = \left(\int_0^n p_i^{1-\sigma} \, di \right)^{1/(1-\sigma)}$ is the price index.

Derivation of Demand from CES (2/2)

Thus, demand for variety i:

$$c_i = \frac{e_i}{p_i} = p_i^{-\sigma} P^{\sigma-1} Y$$

For a single firm, taking P and Y as given (large n), the perceived demand is:

$$q_i = p_i^{-\sigma} K$$

where $K = P^{\sigma-1} Y$ is treated as constant.
Elasticity of demand: $\eta = -\frac{d \ln q_i}{d \ln p_i} = \sigma$.

Firm Technology and Behavior

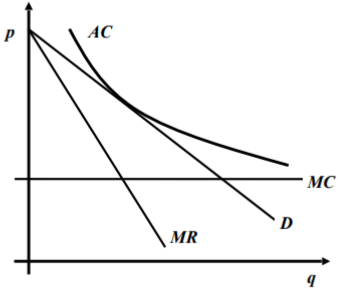
Each firm produces a unique variety with increasing returns:

$$l = f + c \cdot q$$

where l is labor input, q is output, f is fixed cost, c is marginal cost.

- Average cost decreases with q.
- Firms face demand with elasticity σ, set price to maximize profit.
- Wage (w) is normalized to 1. $MC = cw = c$
- Productivity is normalized to 1.

Graphical Illustration of Monopoly Pricing



- The firm sets output q^* where $MR = MC$.
- It sets price p^* on the demand curve corresponding to q^*.
- In the long run, Average Cost (AC) must be tangent to the demand curve at q^* (Zero Profit condition).

Monopolistic Pricing Derivation

- **Profit Maximization:** Firms maximize $\pi = pq - cq - f$ subject to demand $q = p^{-\sigma} K$.
- **Marginal Revenue (MR):** In the monopolistic competitive setting, $MR = p\left(1 - \frac{1}{\sigma}\right)$.
- **Optimality (MR = MC):** Setting $MR = MC = c$ (since $w = 1$):

$$p\left(1 - \frac{1}{\sigma}\right) = c \implies p = \frac{\sigma}{\sigma - 1} c$$

- **Price Rule:** Price is a constant markup over marginal cost. The markup is determined by the elasticity of substitution σ.

$$\frac{p - c}{p} = \frac{1}{\sigma}$$

Monopolistic Pricing Derivation (The Hard Way)

Profit: $\pi = pq - cq - f$, with $q = p^{-\sigma} K$.

$$\pi = p \cdot p^{-\sigma} K - c \cdot p^{-\sigma} K - f = Kp^{1-\sigma} - cKp^{-\sigma} - f$$

FOC: $\frac{d\pi}{dp} = K(1-\sigma)p^{-\sigma} + cK\sigma p^{-\sigma-1} = 0$

$$(1-\sigma)p^{-\sigma} + c\sigma p^{-\sigma-1} = 0$$

$$(1-\sigma)p + c\sigma = 0 \implies p(\sigma - 1) = c\sigma \implies p = \frac{\sigma}{\sigma - 1} c$$

Markup over marginal cost: $\frac{p-c}{p} = \frac{1}{\sigma}$.

Autarky Equilibrium: Output and Price

- **Technology:** $l = f + c \cdot q$ (Increasing Returns to Scale).
- **Pricing (PP curve):** $p = \frac{\sigma}{\sigma-1} c$ (Mark-up determined by σ).
- **Zero Profit (ZZ curve):** Free entry drives profit to zero: $\pi = (p - c)q - f = 0$.

Substituting the price rule into the zero profit condition:

$$\left(\frac{\sigma}{\sigma - 1} c - c \right) q = f \implies \frac{c}{\sigma - 1} q = f$$

Firm Output (q):

$$q^A = \frac{f}{c}(\sigma - 1)$$

Firm output is fixed by the technology (f, c) and consumer preferences (σ).

Autarky Equilibrium: Number of Varieties

Labor Market Clearing: The total labor supply L must equal the labor demanded by all firms (n firms):

$$L = n(f + cq)$$

Substituting the fixed output $q^A = \frac{f}{c}(\sigma - 1)$:

$$L = n\left(f + c \cdot \frac{f}{c}(\sigma - 1)\right) = nf(1 + \sigma - 1) = nf\sigma$$

Number of Varieties (n):

$$n^A = \frac{L}{\sigma f}$$

The number of varieties is limited by the labor supply L and the fixed cost f.

Trade Equilibrium and Gains from Trade

- **Integrated Market:** With free trade, the effective market size is $2L$.
- **Firm Output (q):** The zero-profit condition and price rule are unchanged, so $q^T = q^A$.
- **Total Varieties (n_{total}):** Total available labor is $2L$.

$$n^T_{total} = \frac{2L}{\sigma f} = 2n^A$$

- **Trade Pattern:** Each country specializes in n^A varieties and exports half of its production, leading to Intra-Industry Trade.

Welfare Gain (Variety Effect): Utility is proportional to $n^{1/(\sigma-1)}$.

$$U^T \propto \left(n^T_{total}\right)^{1/(\sigma-1)} = \left(2n^A\right)^{1/(\sigma-1)} = 2^{1/(\sigma-1)} U^A$$

Welfare Factor: $2^{1/(\sigma-1)} > 1$. The gains come purely from the increased number of varieties.

Intuition: Why Trade Raises Welfare in Krugman's Model

- **Consumer Gain (Variety):** Consumers gain access to double the number of differentiated varieties ($n^T_{total} = 2n^A$).
- **Efficiency Gain (Scale):** Firms still operate at the optimal scale $q^A = q^T$, but the combined market ensures that this scale is met for more firms overall.
- **Main Conclusion:** Trade allows consumers to maintain scale economies for domestic firms while simultaneously gaining the full range of global variety.
- Even identical countries benefit trade arises from scale and variety, not comparative advantage.

Home Market Effect (Helpman and Krugman)

- **Concept:** In the presence of trade costs (like tariffs or transport), industries characterized by increasing returns to scale and high trade costs tend to concentrate in the country with the larger domestic market.
- **Mechanism:** Locating in the larger market minimizes overall trade costs.
- **Implication:** The country with the larger market becomes a net exporter of that IRS good.
- **Example:** Concentration of the European automobile industry in Germany, the largest single EU market. US dominace in technologies.

Melitz Model (2003): Firm Heterogeneity

- **Extension of Krugman:** Keeps CES preferences, IRS, and monopolistic competition.
- **Key Addition:** Firms are heterogeneous in terms of **productivity** (ϕ), which is unknown until after an irreversible, sunk entry cost (f_e) is paid.
- **Trade Costs:** Introduces both variable (**iceberg** cost τ) and fixed (**export setup** cost f_{ex}) trade costs.

Key Insight: Selection and Reallocation

Trade liberalization forces a **selection effect**: Only the most productive firms can cover the fixed and variable export costs, leading to **intra-industry reallocation** of resources to high-productivity firms and increasing aggregate industry productivity.

Melitz Model: Key Assumptions

- CES preferences: $U = \left[\int_{\omega \in \Omega} q(\omega)^{\frac{\sigma-1}{\sigma}} d\omega\right]^{\frac{\sigma}{\sigma-1}}$, $\sigma > 1$.
- Labor as sole factor, wage $w = 1$, total L.
- Production: $l = f + q/\phi$, ϕ productivity drawn from $G(\phi)$ (e.g., Pareto).
- Entry: Sunk cost $f_e > 0$ to draw ϕ, exit probability δ.
- Trade: Iceberg costs $\tau \geq 1$, fixed export cost $f_{ex} > 0$.
- Symmetric countries.
- Exporting incurs iceberg variable costs ($\tau \geq 1$, meaning you ship τ units for 1 to arrive) and fixed export costs $f_{ex} > 0$
- Profits: Domestic $\pi_d(\phi) = r_d(\phi)/\sigma - f$, export $\pi_x(\phi) = r_x(\phi)/\sigma - f_x$ (with $f_x = \delta f_{ex}$).
- Revenues: $r_d(\phi) = R\left(\frac{P}{\rho\phi}\right)^{\sigma-1}$, $r_x(\phi) = \tau^{1-\sigma} R\left(\frac{P}{\rho\phi}\right)^{\sigma-1}$.

Melitz Model: Firm Cutoffs

- **Pricing:** Firms set price with the constant mark-up: $p(\phi) = \frac{1}{\rho\phi}$, where $\rho = (\sigma - 1)/\sigma$. More productive firms ($\uparrow \phi$) set lower prices.
- **Domestic Cutoff (ϕ^*):** The minimum productivity required to cover domestic fixed cost f. Firms with $\phi < \phi^*$ exit.

$$\pi_d(\phi^*) = 0$$

- **Export Cutoff (ϕ_x^*):** The minimum productivity required to cover the additional fixed export cost f_{ex} and variable iceberg cost τ.

$$\pi_x(\phi_x^*) = 0 \implies \phi_x^* = \phi^* \cdot \tau \cdot \left(\frac{f_{ex}}{f_d}\right)^{1/(\sigma-1)}$$

- **Result:** Only firms with $\phi \geq \phi_x^* > \phi^*$ export. This explains the empirical regularity that exporters are more productive than non-exporters.

Melitz Model: Equilibrium Conditions

- Free Entry (FE): Expected value of entry equals entry cost: $v = f_e/\delta$, where v is ex-ante firm value.
- Zero Cutoff Profit (ZCP): $\pi(\phi^*) = 0$.
- Aggregate variables: Mass of firms M, price index P, etc., solved via averages over surviving firms.
- In open economy, trade affects cutoffs: Liberalization lowers ϕ^* (more exit), raises average productivity.
- Average profit: $\tilde{\pi} = \int_{\phi^*}^{\infty} \pi(\phi)\mu(\phi)d\phi$, where μ is conditional density.
- For Pareto $G(\phi) = 1 - \phi^{-k}$, $k > \sigma - 1$, cutoffs simplify.
- Labor market: $L = M_e f_e + M(f + f_x \Pr(\text{export})) +$ variable costs.

Melitz Model: Implications of Trade Liberalization

- **Selection Effect:** Lower trade costs (e.g., lower τ or f_{ex}) increase the competition from imports.
  1. The domestic cutoff ϕ^* is pushed u $\implies$ **More low-productivity firms exit** (increased selection).
  2. The export cutoff ϕ_x^* is pushed down $\implies$ **More medium-productivity firms enter the export market**.
- **Aggregate Gains:** Resources reallocate from less-productive exiting firms to more-productive exporting firms, significantly **boosting aggregate productivity and welfare**.
- **Welfare Channels:** Gains come from:
  - Increased variety (Krugman channel).
  - Increased industry-level productivity (Melitz reallocation/selection channel).

Gravity Model of Trade: Empirical Regularity

- **Core Equation (Log-Linear):** Bilateral trade flows (X_{ij}) are proportional to the economic size of countries (Y_i, Y_j) and inversely proportional to the distance (D_{ij}):

$$X_{ij} = G \frac{Y_i^{\alpha} Y_j^{\beta}}{D_{ij}^{\gamma}}$$

  where α, β, γ are positive elasticities (often close to 1).
- **Microfoundations:** Traditional models lacked a rigorous micro-foundation. NTT provides the theoretical structure to derive the gravity equation.
- **Key Insight:** In NTT, the trade cost (τ_{ij}) is a function of distance (D_{ij}), and the trade volume is:

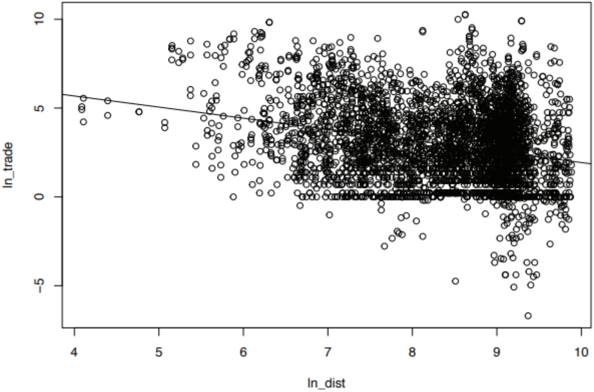
$$X_{ij} \propto \frac{Y_i Y_j}{\tau_{ij}^{\sigma-1}}$$

Gravity Model of Trade: Empirical Regularity

Empirical Regularities

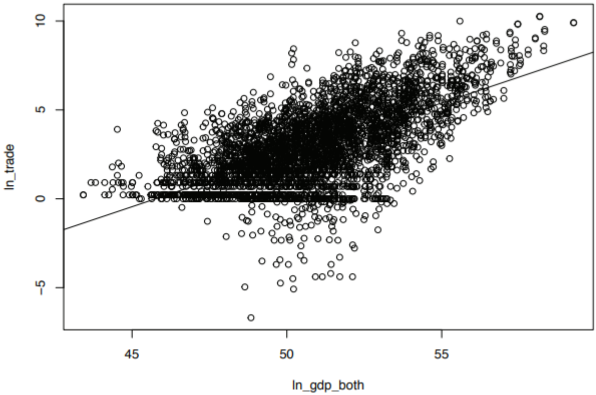
- Trade volume doubles when GDP doubles.
- Trade halves when distance doubles (elasticity $\gamma \approx -1$).

Gravity Relation: Distance and Trade



source: The gravity model of international trade `https://www.unescap.org/sites/default/files/Gravity-model-in-R_1.pdf`

Gravity Relation: GDP and Trade



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Trade Margins: Intensive vs. Extensive (Melitz)

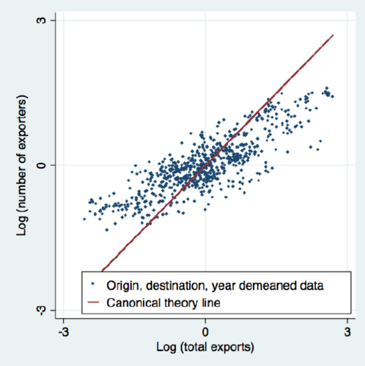
- **Intensive Margin:** The average sales volume per exporting firm. Trade liberalization allows existing exporters to increase sales volumes abroad due to lower variable costs or larger markets. (High-productivity firms expand).
- **Extensive Margin:** The number of exporting firms (or the number of varieties traded). Trade liberalization lowers the export cutoff (ϕ_x^*), allowing more firms to enter export markets.

Impact of Trade Cost Reduction

A reduction in trade costs increases total exports by increasing both the **intensive margin** (more sales per exporter) and the **extensive margin** (more firms/varieties exported).

Intensive Margin: Size of Exporters

- Trade liberalization lowers export thresholds, allowing more firms to enter export markets (increasing the number of exporters). However, it also raises the domestic productivity cutoff, causing low-productivity firms to exit.



source: Fernandes, Ana M., et al. The intensive margin in trade

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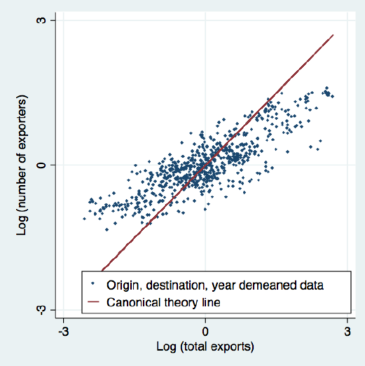


source: Fernandes, Ana M., et al. The intensive margin in trade

Gravity Model Integration: Role in New Trade Theory (1/2)

- New Trade Theory provides theoretical foundations for the gravity equation. (Trade volume increases with economic size and decreases with distance.)
- In Krugman's model with trade costs (e.g., iceberg costs $\tau_{ij} > 1$, where τ_{ij} increases with distance):
- Firms face higher costs to export over distance, reducing trade volumes.
- Bilateral trade derives as $X_{ij} \propto \frac{Y_i Y_j}{\tau_{ij}^{\sigma-1}}$, where σ is elasticity of substitution.
- τ_{ij} often modeled as function of distance, explaining inverse relationship.

Gravity Model Integration: Role in New Trade Theory (2/2)

- Enhances gravity by incorporating trade costs that affect intra-industry trade extent.
- In Melitz model with firm heterogeneity: Gravity holds at aggregate level, but adds extensive marginnumber of exporting firms decreases with distance/costs.
- Trade costs raise export cutoffs, reducing exporters and varieties traded.
- Overall: NTT explains why gravity works even for similar countries and differentiated goods, measuring trade barriers not captured in classical theories.

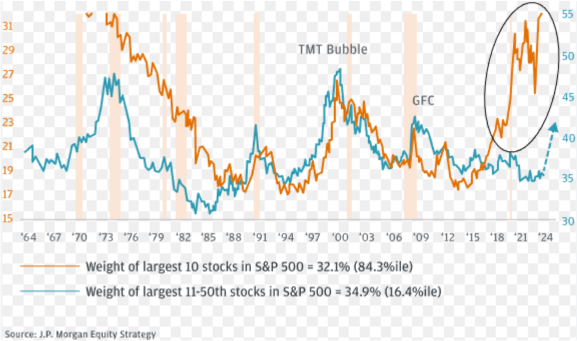
Evidence Supporting New Trade Theory

- High levels of intra-industry trade among developed countries (e.g., cars between Germany and France).
- Growth of trade in services and differentiated goods.
- Studies on firm-level data show productivity gains from exporting.

Market Concentration: Evidence of Increasing Returns?

- NTT and Melitz models predict that economies of scale and fixed costs favor larger, more productive firms and increased concentration.
- **Magnificent 7:** The increasing market share of a few dominant US technology firms provides contemporary evidence of the power of increasing returns (network effects, R&D fixed costs).

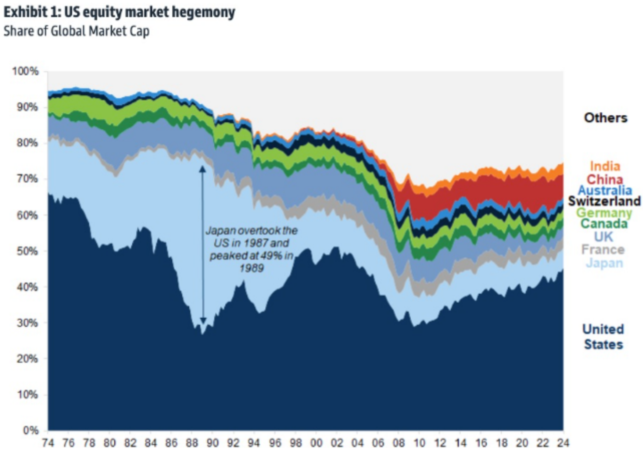
US Stock Market Concentration



source: JP Morgan

Stock Market Concentration

- Share of US stock is increasing



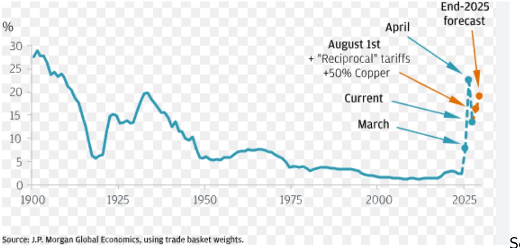
source: Financial Times

Policy and Welfare Implications

- **Welfare Gains:** Trade yields positive welfare gains through:
  - Variety Effect (Krugman).
  - Productivity/Selection Effect (Melitz).
- **Possible losses:** Market concentration and potential monopolies. (Overall, positive effect of trade for similar economies.)
- **Strategic Trade Policy:** NTT suggests a potential justification for government intervention (subsidies) to secure first-mover advantages in IRS-intensive industries (e.g., historical justification for supporting Airbus over Boeing).
- **Protectionism:** Potential for protectionism, but overall trade liberalization benefits consumers through variety. Tariffs reduce competition, lower selection (ϕ^* shifts left), and raise export costs (ϕ_x^* shifts right), potentially reducing aggregate productivity and variety.

US Tariffs

- **Question:** Can tariffs help?
- **NTT View:** Tariffs act as trade costs (τ), reducing both the intensive and extensive margins of trade. While a country may gain from strategic tariffs, the overall global welfare loss is certain.



Source: JP Morgan

Limitations and Extensions

- **Trade Imbalances:** The core Krugman (1980) and Melitz (2003) models assume a static, balanced trade equilibrium (Exports = Imports). They do not explain persistent trade imbalances (e.g., US-China).
- **Need for Integration:** For current account analysis and imbalances, NTT must be integrated with intertemporal models.
- **Extensions:** Subsequent research has incorporated multi-factor production, general equilibrium effects, and political economy into these frameworks.

Summary

- **NTT Complements:** New Trade Theory complements traditional models by explaining modern trade patterns (intra-industry trade).
- **Driving Forces:** Trade is driven by **economies of scale**, **product differentiation**, and **monopolistic competition**.
- **Key Models:** Krugman focuses on variety; Melitz adds **firm heterogeneity** and the **selection/reallocation** effect.
- **Empirical Link:** NTT provides the microeconomic foundation for the robust empirical regularity of the **Gravity Model**.

Sources

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- Treb Allen and Costas Arkolakis lecture notes
- Gianluca Benigno lecture notes (Background Note on Trade and Trade Imbalances)

Thank You

Questions?